

Beyond the Model: A Practitioner's Guide to Harness Engineering

May 26, 2026

Welcome & thank you for joining!



The webinar will run for 45 minutes



Submit Q&A throughout using the Q&A panel



A recording will be sent to all registrants



Meet Our Hosts



Cornelia Davis
Principal Developer Advocate
Temporal



Naresh Rajendra Shah
AI Architect & Advisor (CTO Org)
Grid Dynamics



Today's Agenda

- 00 Intro
- 01 What is an Agent Harness
- 02 Durable Execution as a Part of the Harness
- 03 Configuring the Agent Harness
- 03 From ReAct to CodeAct
- 04 Q&A



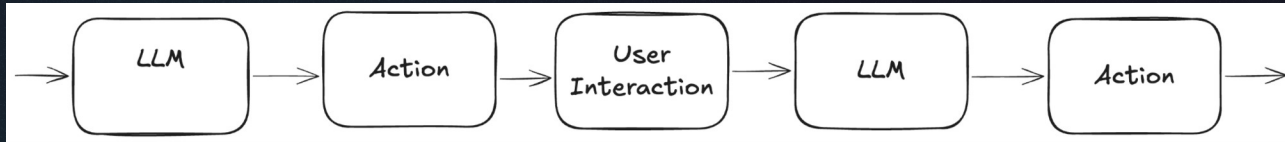
Agent = Model + Harness

That is, the harness is everything in the agent except for the model.

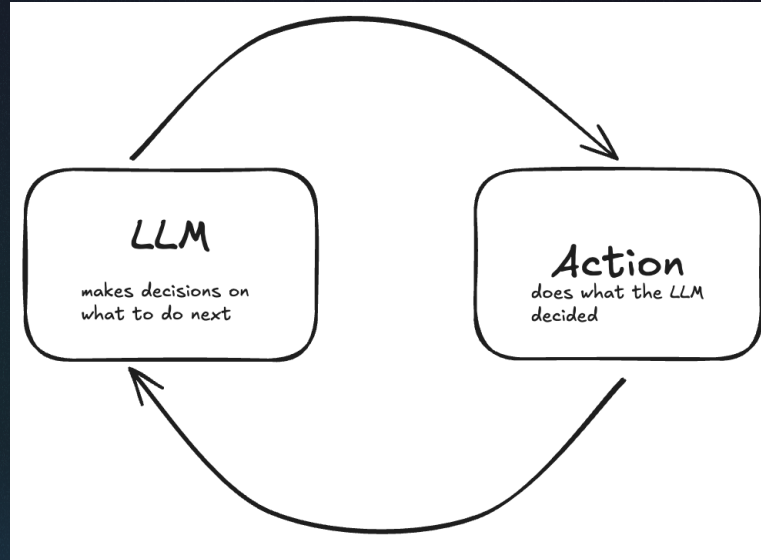


Fixed Flow - Is this an Agent?

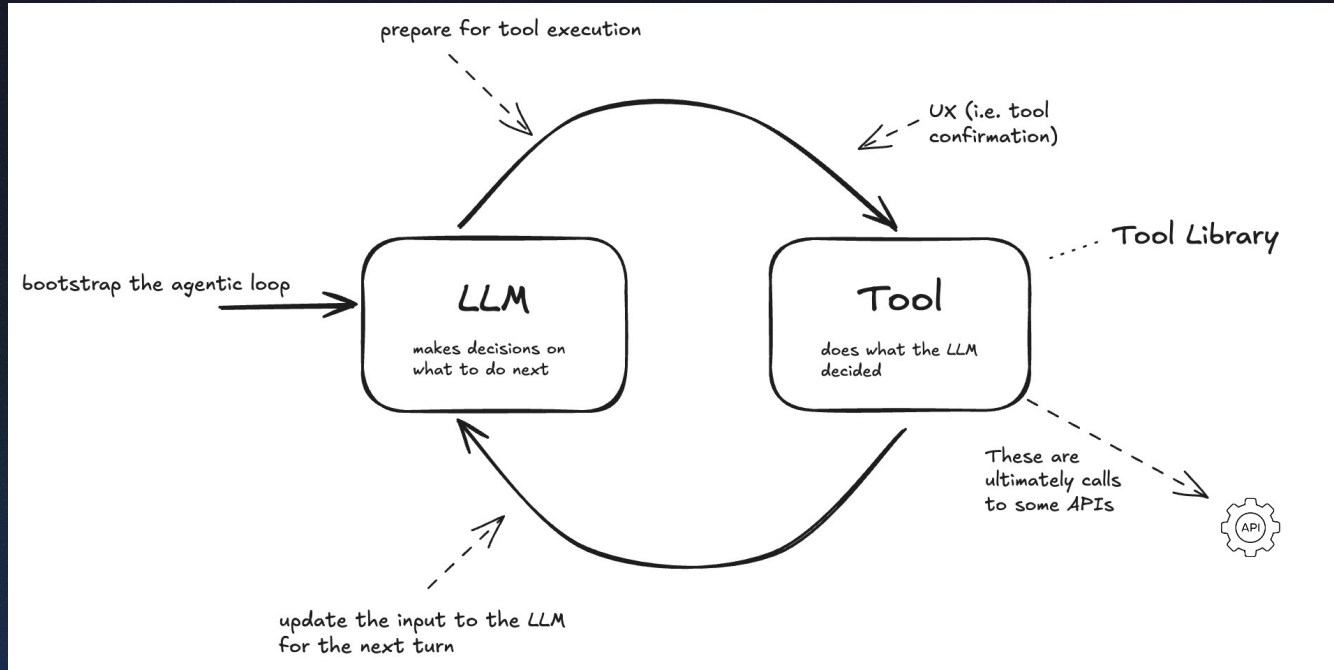
**May
be**



An Agent is one where the LLM has Agency



There is a lot that happens around the LLM



Simple **Agent Demo**

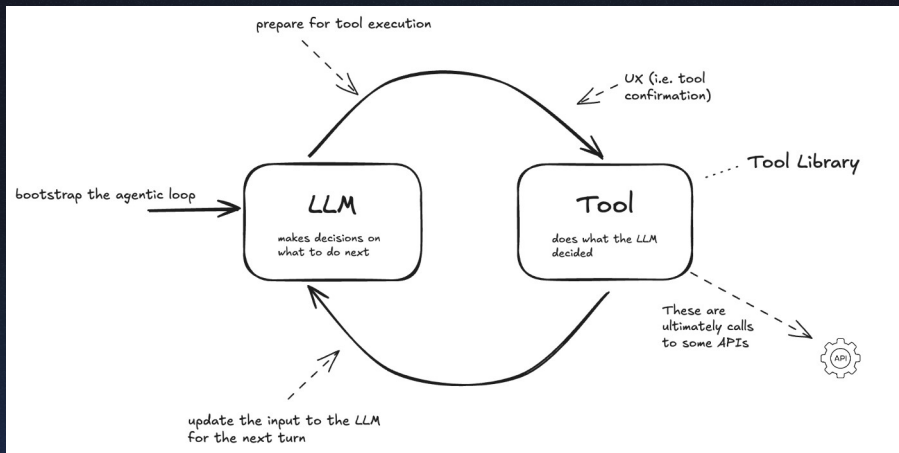


Harness Engineering

It is the engineering all of the things around the model.



What else goes into an **agent harness**?



- Context Engineering
 - System prompts
 - RAG
 - Conversation history management
 - Other guidance 🤖
- Evals/Guardrails
 - Running tests
 - Linting ...
- Agent Skills - a bit of a hybrid...
- and...



Durable Execution

Essential part of the agentic harness



What's Temporal?

- A Durable Execution platform for building reliable, long-running applications.
- By default, Temporal makes workflows fault-tolerant and stateful.
- You write workflows as code, and Temporal ensures they run reliably at any scale.
- Used by companies like
 - Block to ensure that high stakes financial workflows run without incident.
 - OpenAI to orchestrate the process of creating images in ChatGPT





“Temporal is used for **asynchronous workflows and operations inside OpenAI**. Temporal is a neat workflow solution that makes multi-step workflows reliable even when individual steps crash, without much effort by developers. **It’s particularly useful for longer-running workflows like image generation at scale.**”

Gergely Orosz

“Building, launching, and scaling ChatGPT Images”

newsletter.pragmaticengineer.com



Durable Agent Demo



AI Integrations & Partnerships

Agents SDKs

- OpenAI Agents SDK 
- Pydantic AI 
- Scale AI Agentex 
- Vercel AI SDK 
- AWS Strands 
- Google ADK 
- Spring AI
- more in progress

Observability/ Evals Integrations

- Open Telemetry  
- Braintrust  
- Langfuse 
- more in progress

Other

- more in progress



Harness Engineering Applied

(It's not just for coding agents)



Who is Grid Dynamics

- Grid Dynamics (NASDAQ: GDYN) is a Silicon Valley-headquartered, AI-first digital engineering and IT consulting firm building Agentic AI platforms at Fortune 100 companies using Temporal as the underlying orchestration and state management layer.



Harness Engineering Model

The model sits at the center. Everything else — from input guards to durable state — is harness engineering.

USER LAYER	ENTRY & EXIT
GUARDRAILS	SAFETY & VALIDATION
ORCHESTRATION	THE CORE LOOP
CONTEXT GRAPH	DISCOVERY & RELATIONS
CONTEXT	MEMORY & RETRIEVAL
TOOLS	ACTIONS & APIS
OBSERVABILITY	TRACING & METRICS
PERSISTENCE	STATE & DURABILITY



Patterns within AI Harnesses



FIXED TOPOLOGY

Workflows

DAG of nodes drawn at build time

Handles 5 use cases — the 6th breaks it

Process changes = refactor infrastructure & code

SAFE BUT LIMITED

ReAct

Pick from a predefined tool set

Auditable and governable

Agent can never do what you didn't anticipate or tool for

POWERFUL BUT RISKY

CodeAct

Code is the action — entire language available

Composes skills, calls LLMs for subtasks

Hard to audit, hard to govern at scale

THE DEFAULT

Hybrid

Typed tools for governance by default

Bash/code escape hatch when needed

Temporal Skills make every escape hatch durable

Governance by default, CodeAct when needed. *Temporal Skills close the gap — each is a durable, sandboxed Activity.*



Simple Harness

Demo



Agent uses Search tool, then composes a nice html

Durable execution, source-backed claims, visible gaps, and human approval before use.

BUSINESS NEED

The Problem

Marketing needs
fresh customer-
proof content
without manual
research
bottlenecks or
uncited claims

AGENT TASK

What It Does

Search [Temporal.io](https://temporal.io)
website case-study
pages, extract proof
points, and
compose a cited
marketing page

TEMPORAL VALUE

Why Temporal

Durable workflow:
retries, state
tracking, inspection,
approval gates, and
safe partial results

*At runtime, not just
when building the
workflow*

DEMO PAYOFF

What You See

ReAct shows
flexible step-by-step
research. CodeAct
shows generated
workflow code with
stronger structure

TAKEAWAY

The Point

A safer pattern for
turning live web
research into
reviewable
business artifacts

This is not just "search plus nice HTML." *It is governed content automation — durable, cited, and human-approved.*



A Research Agent



THREE WAYS THE AGENT CAN WORK

SIMPLE

Prompt in, text out

LLM does everything
in one shot

no tools, no code

REACT

Think → Act → Observe

LLM calls tools in a loop
web search, extract, analyze

tools predefined by engineers

CODEACT

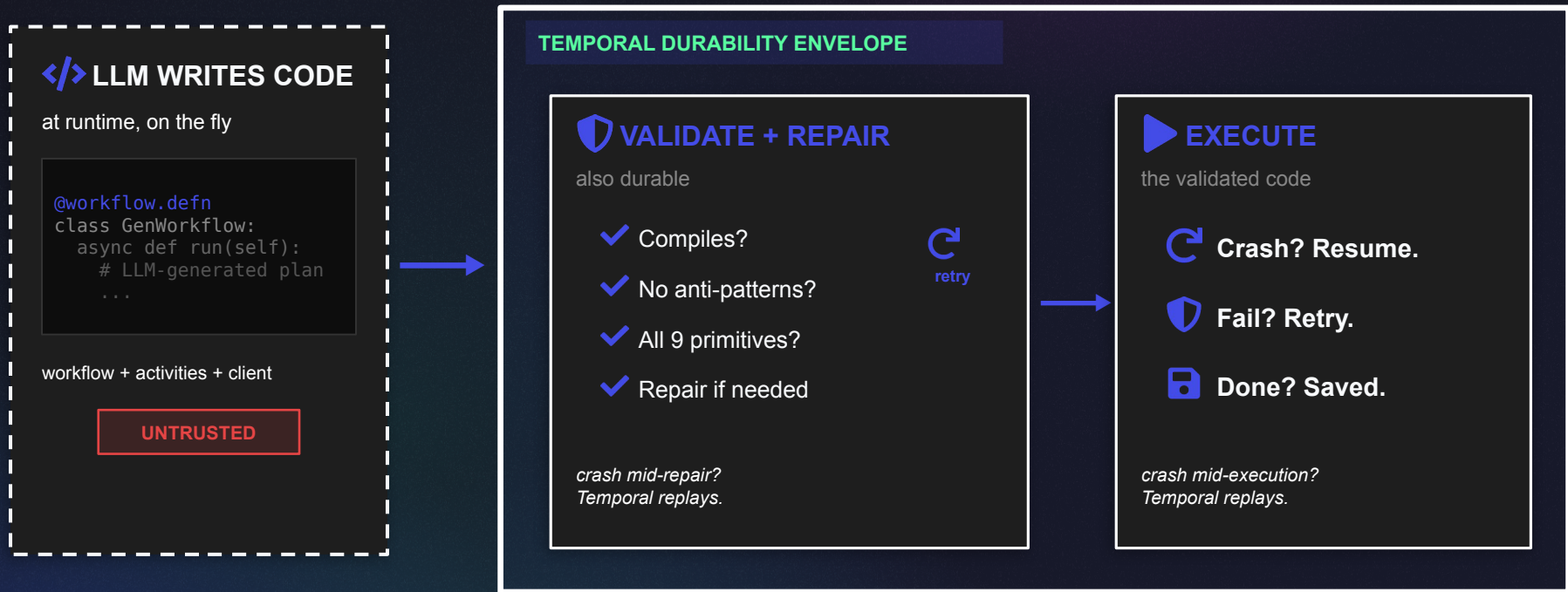
LLM writes a workflow

Generates Temporal code
at runtime — then runs it

Tools built by agents themselves



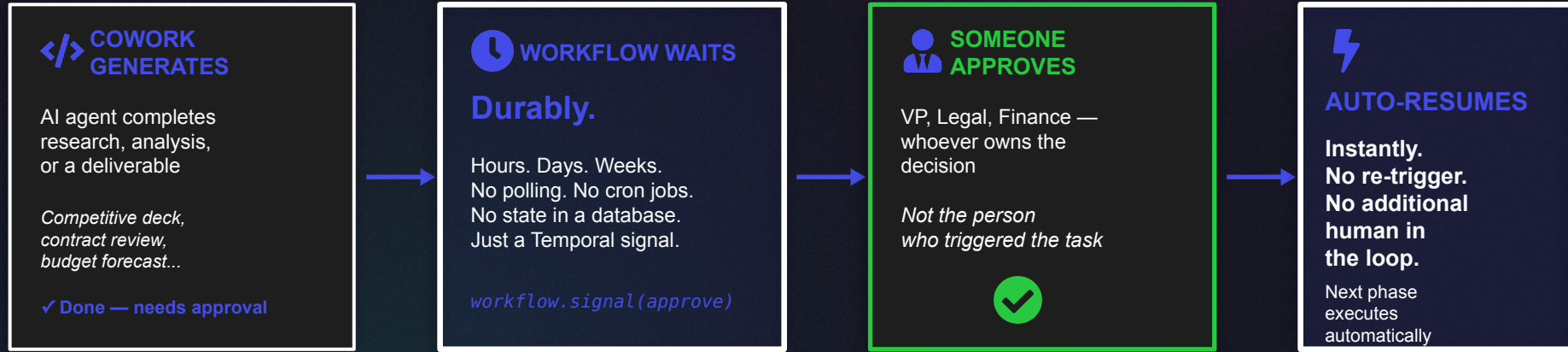
Zooming into CodeAct



The LLM wrote the code. Temporal made everything durable — even the validation.



In Real Business World - You are not alone



Durability isn't just crash recovery.
It's the ability to wait — for days — and pick up exactly where you left off.

Temporal signals turn approval workflows from fragile state machines into durable conversations



What we covered

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CTA



Q&A



